Theorem: invertible matrix equivalency

Theorem 7.19 (Invertible Matrix Theorem)**.** *Let A be an $m \times n$ matrix and $T : V \to W$ a linear transformation such that $[T]_{C \leftarrow B} = A$ for bases B of V and C of W. Then the following statements are all equivalent.*

(a) *A is invertible.*

(b) *For every $\mathbf{b} \in \mathbb{R}^n$, $A\mathbf{x} = \mathbf{b}$ has a unique solution.*

(c) *$A\mathbf{x} = \mathbf{0}$ has only the trivial solution.*

(d) *The reduced row echelon form of A is I_n.*

(e) *A can be expressed as a product of elementary matrices.*

(f) $\text{rank}(A) = n$.

(g) $\text{nullity}(A) = 0$.

(h) *The column vectors of A are linearly independent.*

(i) *The column vectors of A span $\mathbb{R}^n$.*

(j) *The column vectors of A form a basis for $\mathbb{R}^n$.*

(k) *The row vectors of A are linearly independent.*

(l) *The row vectors of A span $\mathbb{R}^n$.*

(m) *The row vectors of A form a basis for $\mathbb{R}^n$.*

(n) $\det(A) \neq 0$.

(o) *0 is not an eigenvalue of A.*

(p) *T is invertible.*

(q) *T is one-to-one (injective).*

(r) *T is onto (surjective).*

(s) $\ker(T) = \{\mathbf{0}\}$.

(t) $\text{range}(T) = W$.

(u) *0 is not a singular value of A.*

Proof. We prove the cycle

$$(a) \Rightarrow (b) \Rightarrow (c) \Rightarrow (d) \Rightarrow (e) \Rightarrow (a),$$

then use established linear-algebra identities to collect the remaining equivalences.

(a) ⇒ (b). Suppose A is invertible. For any $\mathbf{b} \in \mathbb{R}^n$, the vector $\mathbf{x} = A^{-1}\mathbf{b}$ is a solution of $A\mathbf{x} = \mathbf{b}$. If $\mathbf{x}'$ is another solution, then

$$A\mathbf{x}' = \mathbf{b} \implies \mathbf{x}' = A^{-1}A\mathbf{x}' = A^{-1}\mathbf{b} = \mathbf{x}.$$

Hence the solution is unique.

(b) ⇒ (c). Taking $\mathbf{b} = \mathbf{0}$, statement (b) gives a unique solution to $A\mathbf{x} = \mathbf{0}$. Since $\mathbf{x} = \mathbf{0}$ is always a solution, it must be the unique one.

(c) ⇒ (d). The homogeneous system $A\mathbf{x} = \mathbf{0}$ has only the trivial solution iff the system has no free variables, i.e., every column of A contains a pivot. Since A is $n \times n$, there are exactly n pivots, one in each column and each row, so the reduced row echelon form is I_n.

(d) ⇒ (e). Row reduction to I_n amounts to multiplying A on the left by a sequence of elementary matrices $E_1, \ldots, E_k$:

$$E_k \cdots E_1 A = I_n.$$

Since elementary matrices are invertible,

$$A = E_1^{-1} \cdots E_k^{-1},$$

which is a product of elementary matrices (each E_i^{-1} is itself elementary).

(e) ⇒ (a). Elementary matrices are invertible, and a product of invertible matrices is invertible, so A is invertible.

This closes the main cycle. We now establish the remaining equivalences.

(a) ⇔ (f). By the Rank–Nullity Theorem, $\operatorname{rank}(A) + \operatorname{nullity}(A) = n$. A invertible ⇔ (c) ⇔ $\operatorname{nullity}(A) = 0 \Leftrightarrow \operatorname{rank}(A) = n$.

(a) ⇔ (g). $\operatorname{nullity}(A) = 0 \Leftrightarrow A\mathbf{x} = \mathbf{0}$ has only the trivial solution ⇔ (c) ⇔ (a).

(a) ⇔ (h). The columns of A are linearly independent iff $A\mathbf{x} = \mathbf{0}$ has only the trivial solution (c), which is equivalent to (a).

(a) ⇔ (i). $\operatorname{rank}(A) = n$ iff the column space of A equals $\mathbb{R}^n$, i.e., the columns span $\mathbb{R}^n$. This is (f), equivalent to (a).

(a) ⇔ (j). The columns form a basis for $\mathbb{R}^n$ iff they are linearly independent (h) and span $\mathbb{R}^n$ (i), both equivalent to (a).

(a) ⇔ (k). A is invertible iff A^T is invertible iff the columns of A^T are linearly independent iff the rows of A are linearly independent.

(a) ⇔ (l). A invertible ⇒ $\operatorname{rank}(A^T) = n$ ⇒ the row space of A equals $\mathbb{R}^n$, i.e., the rows span $\mathbb{R}^n$. The converse is symmetric.

(a) ⇔ (m). Combining (k) and (l): rows form a basis iff they are linearly independent and span $\mathbb{R}^n$, both equivalent to (a).

(a) ⇔ (n). A is invertible iff $\det(A) \neq 0$. (Standard determinant characterization of invertibility.)

(a) $\Leftrightarrow$ (o). 0 is an eigenvalue of A iff $A\mathbf{x} = 0 \cdot \mathbf{x} = \mathbf{0}$ has a nontrivial solution iff A is not invertible. Contrapositively, A invertible $\Leftrightarrow$ 0 is not an eigenvalue.

(a) $\Leftrightarrow$ (p). Since $[T]_{C \leftarrow B} = A$, the linear transformation T is invertible iff its matrix representation A is invertible.

(a) $\Leftrightarrow$ (q). T is injective iff $\ker(T) = \{\mathbf{0}\}$ iff $A\mathbf{x} = \mathbf{0}$ has only the trivial solution iff (c) iff (a).

(a) $\Leftrightarrow$ (r). T is surjective iff $\text{range}(T) = W$ iff $\text{rank}(A) = n$ iff (f) iff (a).

(a) $\Leftrightarrow$ (s). $\ker(T) = \{\mathbf{0}\}$ iff T is injective (q) iff (a).

(a) $\Leftrightarrow$ (t). $\text{range}(T) = W$ iff T is surjective (r) iff (a).

(a) $\Leftrightarrow$ (u). By definition, 0 is a singular value of A iff 0 is an eigenvalue of $A^T A$.

(a)$\Rightarrow$(u): If A is invertible, then A^T is also invertible, so $A^T A$ is invertible, and thus 0 is not an eigenvalue of $A^T A$, i.e., 0 is not a singular value of A.

(u)$\Rightarrow$(a): If 0 is not a singular value of A, then 0 is not an eigenvalue of $A^T A$, so $A^T A$ is invertible by (o). By Theorem 3.28 (or the standard result that $A^T A$ invertible $\Rightarrow$ A invertible), A is invertible.

Hence all 21 statements are mutually equivalent. $\square$